

Quantum Mechanics III

Lecture 2

Generalized Measurements and Instruments

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$E_k \geq 0, \sum_k E_k = I$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$p_k = \text{Tr}[\mathcal{M}_k(\rho)]$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$\rho_k = \mathcal{M}_k(\rho)/p_k$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 2; Wiseman selections 1–2; QDC detector block